

Table 1: STR allele frequencies (Dutch Caucasians n=231 individuals)

D3S1358	Dutch C	vWA	Dutch C	D16	Dutch C
12	0.000	11	0.000	8	0.015
13	0.002	13	0.000	9	0.123
14	0.091	14	0.067	10	0.067
15	0.281	15	0.076	11	0.340
16	0.253	16	0.203	12	0.279
17	0.193	17	0.303	13	0.162
18	0.167	18	0.223	13.3	0.000
19	0.011	19	0.110	14	0.013
20	0.002	20	0.013	15	0.000
		21	0.004		
FGA	Dutch C	D21S11	Dutch C	D19	Dutch C
18	0.013	27	0.017	9	0.000
18.2	0.000	28	0.180	10	0.002
19	0.058	29	0.223	11	0.004
19.2	0.000	29.2	0.000	12	0.061
20	0.145	30	0.271	12.2	0.000
21	0.177	30.2	0.030	13	0.255
22	0.173	31	0.078	13.2	0.011
22.2	0.015	31.2	0.074	14	0.359
23	0.136	32	0.011	14.2	0.032
23.2	0.006	32.2	0.091	15	0.165
24	0.158	33	0.002	15.2	0.037
25	0.074	33.1	0.000	16	0.037
25.2	0.000	33.2	0.022	16.2	0.026
26	0.028	34	0.000	17	0.002
27	0.013	34.2	0.000	17.2	0.006
28	0.002	35	0.000	18.2	0.002
29	0.000	35.2	0.002		
31.2	0.000	36	0.000		
45.2	0.000				

D18S15	Dutch C	D2	Dutch C	D8	Dutch C
9	0.000	15	0.000	8	0.019
10	0.011	16	0.048	9	0.011
11	0.006	17	0.203	10	0.078
12	0.134	18	0.076	11	0.087
13	0.108	19	0.128	12	0.147
14	0.182	20	0.171	13	0.346
15	0.117	21	0.015	14	0.180
16	0.152	22	0.030	15	0.102
17	0.141	23	0.097	16	0.028
18	0.071	24	0.095	17	0.002
19	0.039	25	0.117	18	0.000
20	0.026	26	0.017		
21	0.006	27	0.002		
22	0.004				
23	0.000				
25	0.002				
THO	Dutch C	TPOX	Dutch C	CSF	Dutch C
5	0.006	6	0.000	7	0.002
6	0.225	7	0.000	8	0.000
7	0.219	8	0.555	9	0.025
8	0.104	9	0.070	10	0.241
9	0.132	10	0.047	11	0.291
9.3	0.307	11	0.264	12	0.358
10	0.006	12	0.065	13	0.077
				14	0.005
D7	Dutch C	D5	Dutch C	D13	Dutch C
7	0.027	7	0.000	7	0.002
8	0.152	8	0.005	8	0.119
9	0.167	9	0.037	9	0.085
10	0.261	10	0.052	10	0.060
11	0.194	11	0.388	11	0.286
12	0.154	12	0.388	12	0.294
13	0.040	13	0.127	13	0.109
14	0.005	14	0.002	14	0.045
		16	0.000	15	0.000

Situation: sample matches victim and suspect.

Condition: two contributors, victim is contributor (internal sample).

Crime Sample	Complainant	Suspect	LR
A_1, A_2, A_3, A_4	A_1, A_2	A_3, A_4	$\frac{1}{2p_3p_4}$
A_1, A_2, A_3	A_1, A_2	A_1, A_3 or A_2, A_3 or A_3, A_3	$\frac{1}{p_3(2p_1+2p_2+p_3)}$
A_1, A_2, A_3	A_1, A_1	A_2, A_3	$\frac{1}{2p_2p_3}$
A_1, A_2	A_1, A_2	A_1, A_1 or A_1, A_2 or A_2, A_2	$\frac{1}{(p_1+p_2)^2}$
A_1, A_2	A_1, A_1	A_1, A_2 or A_2, A_2	$\frac{1}{p_2(2p_1+p_2)}$
A_1, A_1	A_1, A_1	A_1, A_1	$\frac{1}{p_1^2}$

Situation: sample matches suspect 1.

Condition: two contributors.

Crime Sample	Known Suspect	LR
A_1, A_2, A_3, A_4	A_1, A_2	$\frac{1}{12p_1p_2}$
A_1, A_2, A_3	A_1, A_2	$\frac{1}{2p_1+2p_2+p_3}$
A_1, A_2, A_3	A_1, A_1	$\frac{1}{12p_1p_2(p_1+p_2+p_3)}$
A_1, A_2	A_1, A_2	$\frac{1}{6p_1(p_1+p_2+p_3)}$
A_1, A_2	A_1, A_1	$\frac{(p_1+p_2)^2}{2p_1p_2(2p_1^2+3p_1p_2+2p_2^2)}$
A_1, A_1	A_1, A_1	$\frac{2p_1+p_2}{2p_1(2p_1^2+3p_1p_2+2p_2^2)}$
A_1, A_1	A_1, A_1	$\frac{1}{p_1^2}$

Situation: sample matches suspect 1 and suspect 2.

Condition: two contributors.

Crime Sample	Suspect 1 , Suspect 2	LR
A_1, A_2, A_3, A_4	Any combination of alleles	$\frac{1}{24p_1p_2p_3p_4}$
A_1, A_2, A_3	that contains the	$\frac{1}{12p_1p_2p_3(p_1+p_2+p_3)}$
A_1, A_2	alleles in the crime sample	$\frac{1}{2p_1p_2(2p_1^2+3p_1p_2+2p_2^2)}$
A_1		$\frac{1}{p_1^4}$

Parent Index

Parent	Alleged Child	PI
aa	aa	$\frac{1}{p_a}$
aa	ab	$\frac{1}{2p_a}$
ab	aa	$\frac{1}{2p_a}$
ab	ab	$\frac{p_a + p_b}{4p_a p_b}$
ab	ac	$\frac{1}{4p_a}$

Sibling Index

Person 1	Person 2	SI
aa	aa	$\frac{(1+p_a)^2}{4p_a^2}$
aa	ab	$\frac{1+p_a}{4p_a}$
aa	bb or bc	$\frac{1}{4}$
ab	aa	$\frac{1+p_a}{4p_a}$
ab	ab	$\frac{1+p_a+p_b+2p_a p_b}{8p_a p_b}$
ab	ac	$\frac{1+2p_a}{8p_a}$
ab	cc	$\frac{1}{4}$
ab	cd	$\frac{1}{4}$

Formulas for θ correction

The formulas for the genotype frequencies based on the allele frequencies are given by

$$P_{ii} = \frac{(3\theta + (1 - \theta)p_i)(2\theta + (1 - \theta)p_i)}{(1 + \theta)(1 + 2\theta)}, \quad (2)$$

$$P_{ij} = \frac{2(\theta + (1 - \theta)p_i)(\theta + (1 - \theta)p_j)}{(1 + \theta)(1 + 2\theta)}. \quad (3)$$